

MASTER'S COMPREHENSIVE EXMINATION

March 24, 2007

Part II

Applied (90 minutes)

Instructions: Answer all three questions. Open book and open notes.

- 1) A study was conducted on a random sample of 50 male and 50 female college students. Each student was asked "If you had to choose one activity from the following, which would you choose? playing computer games, playing sports, or talking with friends." The data was as follows:

	<u>Male</u>	<u>Female</u>	<u>Total</u>
Computer games	20	15	35
Talking with friends	14	23	37
Sports	<u>16</u>	<u>12</u>	<u>28</u>
Total	50	50	100

- a) Test ($\alpha=0.05$) whether there is a difference between the first choices of males and of females. (The Null hypothesis is no difference; the alternative hypothesis is some difference.)
- b) Just looking at the 65 students whose first choice was talking with friends or sports, test whether there is a difference between males and females. (The Null hypothesis is no difference, the alternative hypothesis is that males choose sports more often. Give approximate p-value.)